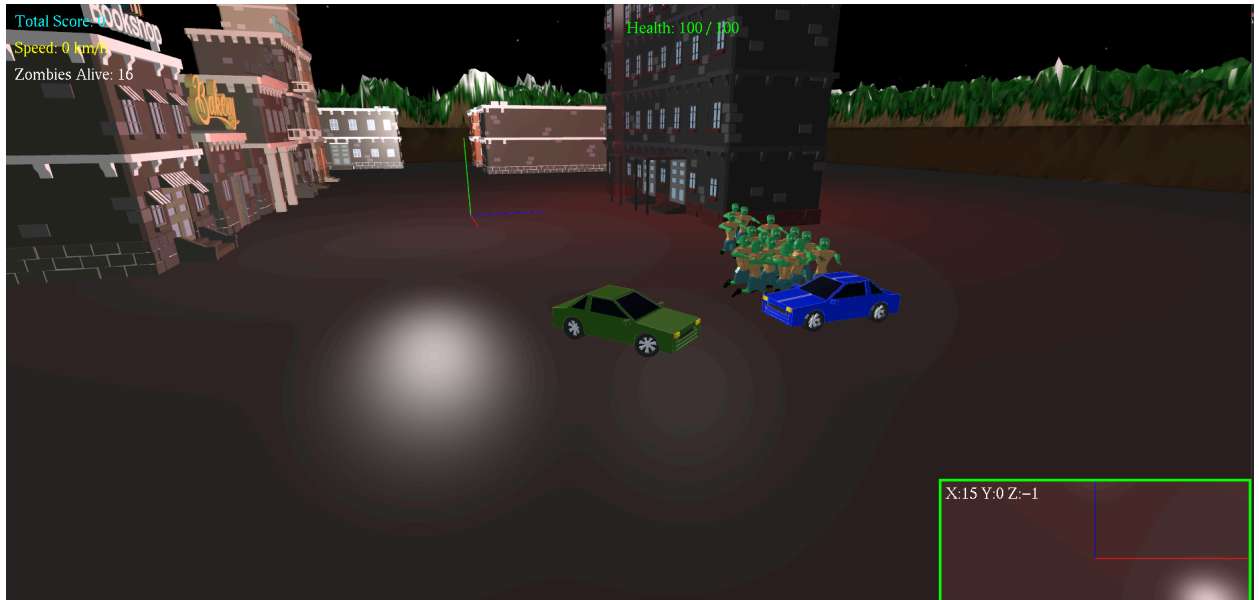
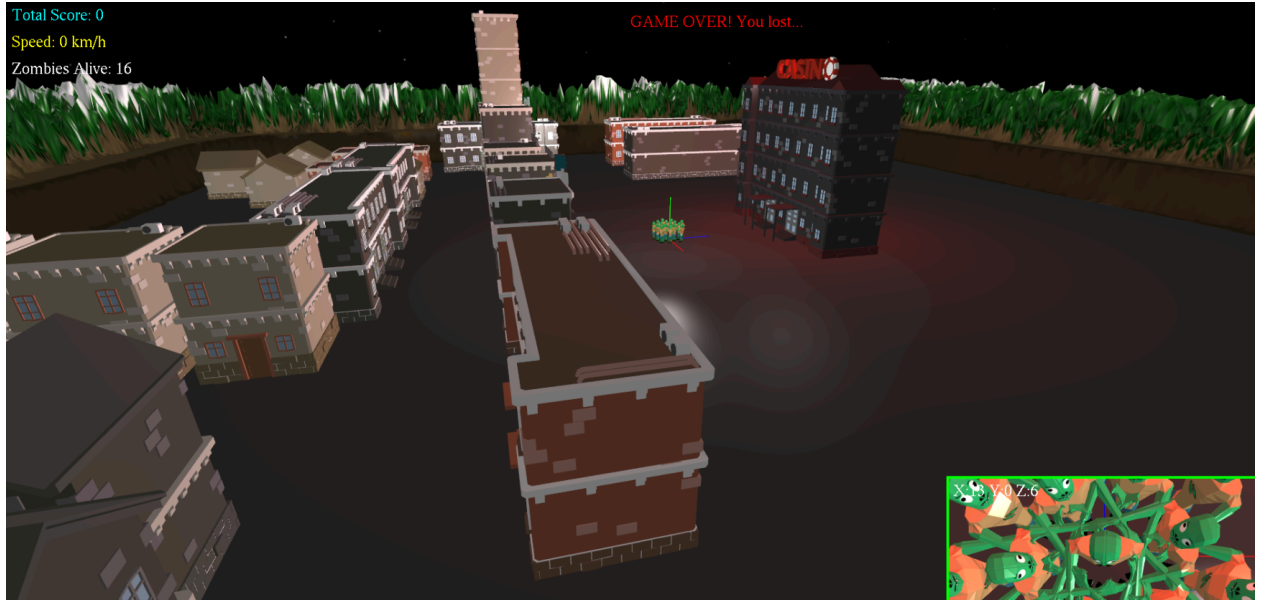


1. Brain Bumpers by *Daniel Ivanilov* and *Maksim Shrkabak*, in Section 01, Spring 2026

2. Gameplay:





3. Once you have the main folder open containing Client and Server folders, you will go to Client, **run buildTAGE.bat** and open up run.bat in notepad or some equivalent, and set the **isMultiplayerMode** to true or false, depending on if you want to play in multiplayer or not. This game has NPCs server sided, so they will not exist in singleplayer, but it is still possible to play in both modes. If you have set isMultiplayerMode = true and keep the default port, then you will save that file to update it. **If you changed the port** with multiplayer mode **true**, you must reflect this port in the **Server**. Now you will go into the Server folder, **run buildTAGE.bat** and if you changed the port on the Client/run.bat, you will update this in the Server/run.bat. Once you have all the information updated, you will **run the server first** → **Server/compile_and_run.bat**, then run the client after → **Client/build_and_run.bat**. Both the Server and Client should be compiled and running now.
4. Upon loading into the game, if the server has not already started the game with others playing, you will be put into a “lobby”. Here, you will be able to cycle and change your avatar to a few of the options pressing **Q** or **E**. Once you are satisfied with your avatar skin, you can **press R to ready up**. The game will not start until **all** connected parties are also readied up. Upon starting the game, you will spawn into the game, in which it will spawn a massive wave of zombies, and your goal is to run over as many as possible

without losing your life, increasing your score for every zombie you run over. If you end up dying, the game will be over for you, unless you scored the most points.

5. Controls for every action with inputs: Keyboard + Mouse → Gamepad.

a. Cycling Avatar Skin (Multiplayer limited to pre-game start):

- i. Keyboard → Q / E
- ii. Gamepad → NA

b. Toggle Ready / Unready (Multiplayer ONLY):

- i. Keyboard → R
- ii. Gamepad → B

c. Disconnect from Server and close game (Multiplayer ONLY):

- i. Keyboard → Z
- ii. Gamepad → NA

d. Moving Forward, Back, and Turning Left & Right:

- i. Keyboard → W / A / S / D
- ii. Gamepad → Left Stick (Y-Axis) & Left Stick (X-Axis)

e. Looking Around:

- i. Mouse → Move any Direction
- ii. Gamepad → Right Stick (X-Axis) & Right Stick (Y-Axis)

f. Zooming In/Out

- i. Mouse → Scroll Wheel UP / DOWN
- ii. Gamepad → RB/LB

g. Toggling Avatar Headlights:

- i. Keyboard → L
- ii. Gamepad → Y

h. Toggle Axis at center:

- i. Keyboard → X
- ii. Gamepad → X

i. Physics world wireframe:

- i. Keyboard → SPACEBAR
- ii. Gamepad → NA

6. Currently, our game has a few **Lights + Global Ambient**.
 - a. Ambient is used to set the game's setting.
 - b. We have a couple Spotlights used as the Avatar's Headlights, which you can turn **on/off** pressing L if using Keyboard(refer to 5g for specifics).
 - c. Red positional light near the casino.

7. Many custom messages were added beyond the base framework:

Message	Direction	Purpose
ready	C→S	Player ready-up in lobby
setTex	C→S	Change avatar skin
killNPC	C→S	Player ran over a zombie
dead	C→S	Player reports their own death
isnear	C→S	NPC proximity check
wave	S→C	Announce wave start + zombie count
createNPC	S→C	Spawn zombie ghost on client
updateNPC	S→C	Update zombie position/rotation
removeNPC	S→C	Remove zombie (killed)
scoreUpdate	S→C	Broadcast all player scores
npcAttack	S→C	Notify client an NPC hit them
playerDead	S→C	Notify all clients a player died
startMsg	S→C	Game start announcement
turn	S←→C	Wheel steering angle for

		other players
--	--	---------------

8. TAGE Engine Modifications (NOTE: Some of these are not used and are from the previous assignment):

- GameObject.java: Added globalYaw() to rotate the avatars around the Y-axis
- GameObject.java: Added pitch() to allow complex 3D rotation
- GameObject.java: Added spawnObject(...) static helper method to make object creating easier - handles translation, scaling, and Y-axis rotation in a single call.
- HUDmanager.java: Expanded to support additional HUD slots for game state and other messages
- Camera.java: Added yaw() and pitch() methods to support the orbit camera controller
- CameraOrbit3D.java: Custom orbital camera controller allowing the player to orbit, elevate, and zoom the camera around the avatar independently of its movement and orientation.
- JumpController.java: Custom node controller that causes a target object to bounce.

9. Our game's...

- a. **Genre** is Action/Survival,
- b. **Theme** is Zombie Apocalypse,
- c. **Dimensionality** is in 3D,
- d. **Activities** consist of Driving vehicles, Exploration, Survival, and Combat.

10. In the game, **ALL** of our requirements are visible:

- a. **External Models (Custom) with Textures (Custom)**
 - i. Avatar (Car) visible as your character which you control
 - ii. NPC (Zombie) visible during any on-going wave after the game has started. Limited to Multiplayer ONLY.
- b. **Networked Multi-player**

- i. If isMultiplayerMode=true in Client/run.bat, multiple clients can join if information is up to date, with **Server running first**.
- ii. You will see other players as avatars (cars).
- iii. Interchangeable Avatar textures by pressing **Q / E**. Multiplayer requires the game to not be started in order to switch avatar textures to preserve identity. You can switch in singleplayer all you want.

c. Single-player

- i. In Client/run.bat, set isMultiplayerMode=false to enable singleplayer, then run build_and_run.bat, not needing the Server.

d. Skybox and Terrain

- i. Skybox → Look at the sky.
- ii. Terrain → Look at the ground and past the city (easy if you zoom out).

e. Lights

- i. Global ambient light, white, visible everywhere.
- ii. 2 Avatar Headlights (Spotlights), once toggled by pressing L, visible coming out the Headlights of the Avatar (Car) shining forward.
- iii. Casino positional light, reddish-purplish at the casino.

f. HUD

- i. Visible on the top side of the screen. Score, Speed, Zombies alive, Game status, and your Healthbar, If spectating, lost, or won the game.

g. 3D Sound

- i. Ambient creepy sound, audible from anywhere.
- ii. Car Engine, with increasing pitch the faster you go, audible the closer you are zoomed into the vehicle. Other player avatar engine sounds can be heard directionally.
- iii. When a zombie is run over a splash sound can be heard directionally.
- iv. The casino building's location will play a sound on repeat. This sound is directional.

h. Hierarchical SceneGraph

- i. The tires, all four, on the avatar are each a child of the avatar. They update with the parent.

- ii. The signs on top of various buildings, such as the Casino, are children of the building object they reside on top of.

i. Animation

- i. Zombie running animation when they chase any avatar.

j. NPCs

- i. In Multiplayer ONLY since they are server sided, visible as zombies once the game has begun and while a wave is on-going.

k. Physics

- i. Avatars, buildings, and NPCs all have physics. Avatar drives on top of physics terrain, and they can crash into buildings, losing all their speed.
- ii. All different kinds of physics objects are visible after pressing SPACEBAR, enabling all objects' physics and terrain's wireframe.

11. We got all requirements working.

12. Techniques beyond requirements:

- Health system (PlayerHealth.java) - 100 HP, each NPC hit does 10 damage, death triggers spectating
- Spectating mode - on death, camera follows the nearest alive ghost player
- Ready-up lobby - game only starts when all connected players ready up
- Dynamic engine audio pitch - pitch scales with car speed in real time
- Custom CarController - full physics-based driving with speed, turning, and tire rotation

13. Contributions:

a. Daniel:

- i. Worked on networking with ghost avatar and npcs, animation on my zombie NPC, avatar headlight spotlight, importing and placing build models on our "map", NPCs with behavior tree AI, avatar texture switching. Zombie model, rigged and animated. Created additional avatar (car) textures using the initial one by Maksim.

b. Maksim:

- i. Worked on physics, created a car object with tires that rotate based on the direction the player is turning (built CarController). Created and implemented the terrain texture/height map. Imported free to use building objects to build part of the city. Added collision hitboxes to most of the buildings. Implemented game sounds with directional carEngine and casino sounds. Also imported and implemented the ambient sound. Helped set up the ready up status.

Models/assets:

- 1. Car.obj
- 2. LeftTire.obj
- 3. RightTire.obj
- 4. CarTexture1.png
- 5. CityMap.png
- 6. CityTexture.png
- 7. TireTexture.png

14. Custom assets + distributed:

- a. models/Car.obj
 - i. models/LeftTire.obj created by Maksim Shkrabak
 - ii. models/RightTire.obj Created by Maksim Shkrabak
- b. models/Zombie.obj Created by Daniel Ivanilov
- c. animations/zombieMesh.rkm created by Daniel Ivanilov
- d. animations/zombieSkeleton.rks created by Daniel Ivanilov
- e. animations/zombieRunning.rka created by Daniel Ivanilov
- f. textures/CityMap.png created by Maksim Shkrabak
- g. textures/CityTexture.png created by Maksim Shkrabak
- h. textures/CarTexture1.png created by Maksim Shkrabak
- i. textures/CarTexture2.png created by Daniel Ivanilov
- j. textures/CarTexture3.png created by Daniel Ivanilov
- k. textures/TireTexture.png created by Maksim Shkrabak

1. textures/ZombieSkin.png created by Daniel Ivanilov

15. Skybox: xn.jpg, xp.jpg, yn.jpg, yp.jpg, zn.jpg, zp.jpg →

<https://opengameart.org/content/night-sky-skybox-generator>

Sounds: carEngine.wav, casino.wav, splash.wav → <https://pixabay.com/>

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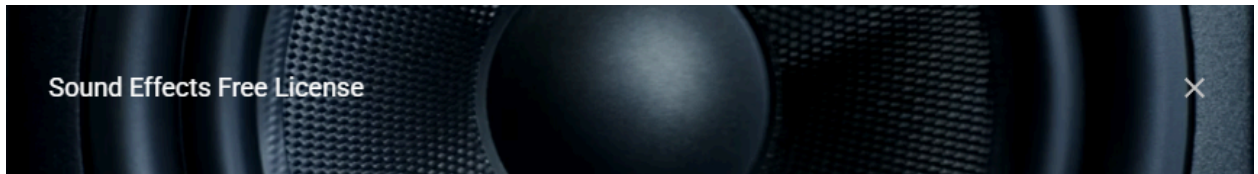
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Textures: CasinoTexture.png, DarkBuildingTexture.png, RedBuildingTexture.png,

SignsTexture.png, YlwBuildingTexture.png →

<https://quaternius.com/packs/ultimatetexturedbuildings.html>

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TwoStoryDouble.obj, TwoStoryHome.obj, TwoStoryRoof.obj, TwoStoryWide.obj →
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